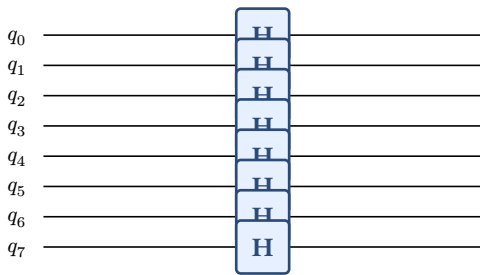
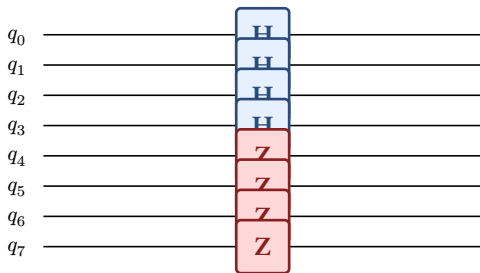


Before training: 8 Hadamards (QFT init)



end-to-end training

After training: 4 H + 4 frozen gates



mixing qubits (superposition, intra-block FFT)

4 frozen qubits = 2^4 block indices

Effective block side: $2^4 = 16$ pixels



Hadamard (mixing)



$\approx \text{diag}(+1, -1)$ (frozen)